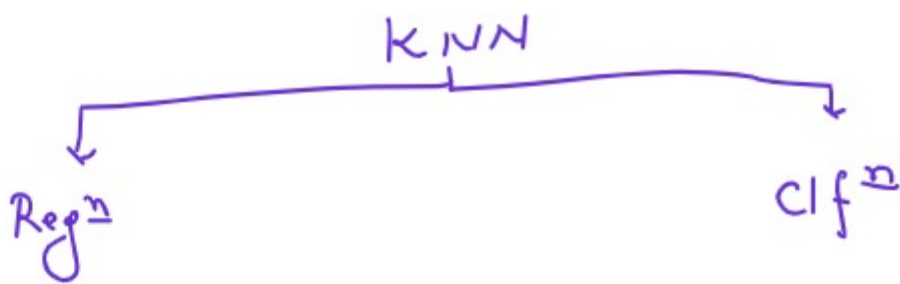
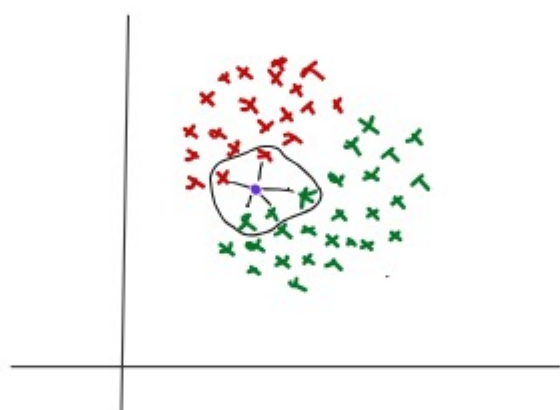


KNN (K-Nearest Neighbour Algorithm)



Agenda

- ① Get Data
- ② Define k Neighbours
- ③ Calculate the Neighbour Distance.
- ④ Assign New instance to Majority of Neighbour.



majority → Red = 2
Green = 3 ✓✓

Distance Formula :

1. EUCLIDIAN DISTANCE

$k = 5$

(Distance formula) $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

2. MANHATTAN DISTANCE

$$D = |x_2 - x_1| + |y_2 - y_1|$$

3. MINKOWSKI DISTANCE

→ Generalization of Both model

$$\left[\sum_{i=1}^n |x_i - y_i|^p \right]^{\frac{1}{p}}$$

∴ $p = 1, p = 2, p = 3, \dots, p = n$